Egocentric Human Demonstrations for a Bimanual Robot
Part I: Three-Cube Stacking

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Short version; Part II covers single-arm approach data. Code, results and sample data are available on request.

***Abstract*—We study how far egocentric demonstrations recorded with a hand-worn UMI-style gripper can be turned into training data for a bimanual robot policy. We collected 349 demonstrations of three-cube stacking in six balanced orders and processed them in two generations. First, MASt3R-SLAM wrist poses with per-episode IMU scale were retargeted to robot joints; a 0.9B X-VLA model trained on the 561 surviving segments predicts the direction of held-out moving human motion (cosine 0.90 at 1.2 s, 39.9 mm error) but predicts spurious motion in static phases and makes no task-useful use of the order instruction. Second, whole episodes were converted to the robot's own relative Cartesian action contract with no feasibility filter (330 episodes, 48,411 training rows). With a shared contract the remaining human-to-robot gap is visual: an ego-only model gives no useful action direction on robot frames, and swapping inputs locates the gap in the wrist images. As an initialization, ego pretraining speeds up early robot fine-tuning, on 312 and on 30 robot episodes alike, but the training-loss advantage disappears (by 60k steps on 312 episodes; by about 100k–200k steps on 30). An ego-initialized model fine-tuned on 312 robot episodes stacked three cubes in closed loop, but trials were not counted, so we report a demonstration, not a success rate.**

I. Introduction

Egocentric human demonstrations are cheaper to collect than robot teleoperation, and hand-worn grippers such as UMI [1] make the human end-effector look like the robot's. Whether such data transfer to a specific robot depends on calibration, metric pose recovery, conversion into the robot's action space and filtering, each of which can silently change what a policy learns. EgoMimic [2] and MimicPlay [3] report closed-loop gains from human video; we instead audit one complete pass through this chain on real data. The question is narrow: *after all processing, how much robot-executable action structure does a pretrained vision-language-action (VLA) model learn from egocentric human demonstrations, and does it help robot fine-tuning?* Evaluation is mostly offline or training-loss based; layouts were not recorded and all six orders appear in training, so we test neither spatial nor task-order generalization. This short version condenses the full report (kept in the repository) and brings run status to 7 October 2026.

II. Data and Poses

Device and task. Each hand wears a HandUMI unit [8] (open source; VR tracking replaced by a wrist IMU): a 1080p30 fisheye wrist camera, a 200 Hz IMU and a jaw encoder; a fixed Logitech C922 records the table. Three 50 mm cubes are stacked on a plate in one of six orders named by the instruction. One operator recorded 349 episodes in 15 sessions on 15–17 September 2026 (2.20 h; median 23.05 s), with the next order chosen as the least collected (51–64 per order), so orders are balanced, not randomized. Cube layouts varied through manual resets but were not recorded (Fig. 1). A later robot-like session (HRL80) added 60 episodes, 10 per order.

Calibration. Kannala–Brandt fisheye intrinsics; Kalibr [4] camera–IMU calibration with time offsets of 30.2 and 32.8 ms and mean reprojection errors of 6.0 and 5.6 px, high enough that pose accuracy is validated separately; a caliper camera-to-TCP transform measured on one unit.

Wrist poses. UMI's per-scene ORB-SLAM3 map failed on our data (median coverage 0.03, frequent reversals). We run MASt3R-SLAM [5] on the RGB video alone and recover metric scale per episode from the IMU. On 14 ArUco benchmark episodes this gives coverage 1.00 versus 0.03, scale error p50/p90 4.6/17.0% (18.5/23.8% for one global constant), rotation error p50 0.84° and a 16-step relative position error p90 of 11.3 mm.

III. Generation 1: Joint-Space Retargeting

Processing. Each TCP trajectory is retargeted by IK to the reBot B601 6-DoF arm, re-anchored at a fixed robot start pose. 65 episodes fail the $\leq$20 ms synchronisation requirement; of 1,306 candidate bimanual 65-frame segments, 677 leave the robot workspace, 50 fail the metric check and 18 fail IK, leaving 561 (43.0%) from 259 episodes, split by episode into 233 train / 26 val. We fine-tune X-VLA [6] (879M parameters, from `xvla-base`, via LeRobot [7]) to predict 30-step chunks of joint displacements from three views, the instruction and a 14-D state, for 300k steps.

Offline results (Table I, Fig. 2). Errors over all samples are small (1.5–5.7 mm) because 77% of samples are near-static. On the 23.4% that move at least 20 mm, the error at k=30 (1.2 s) is 39.9 mm, the direction cosine rises from 0.72 to 0.90, and joint MAE is 13–32% below a zero-motion predictor (6.59 versus 8.68° at k=30). On all samples the policy is *worse* than predicting no motion: it creeps during static phases. **Prompt swap.** The six order instructions move the prediction by 4.8 mm at k=30 (50$\times$ the noise spread, but 8% of the 62 mm predicted displacement), and the true

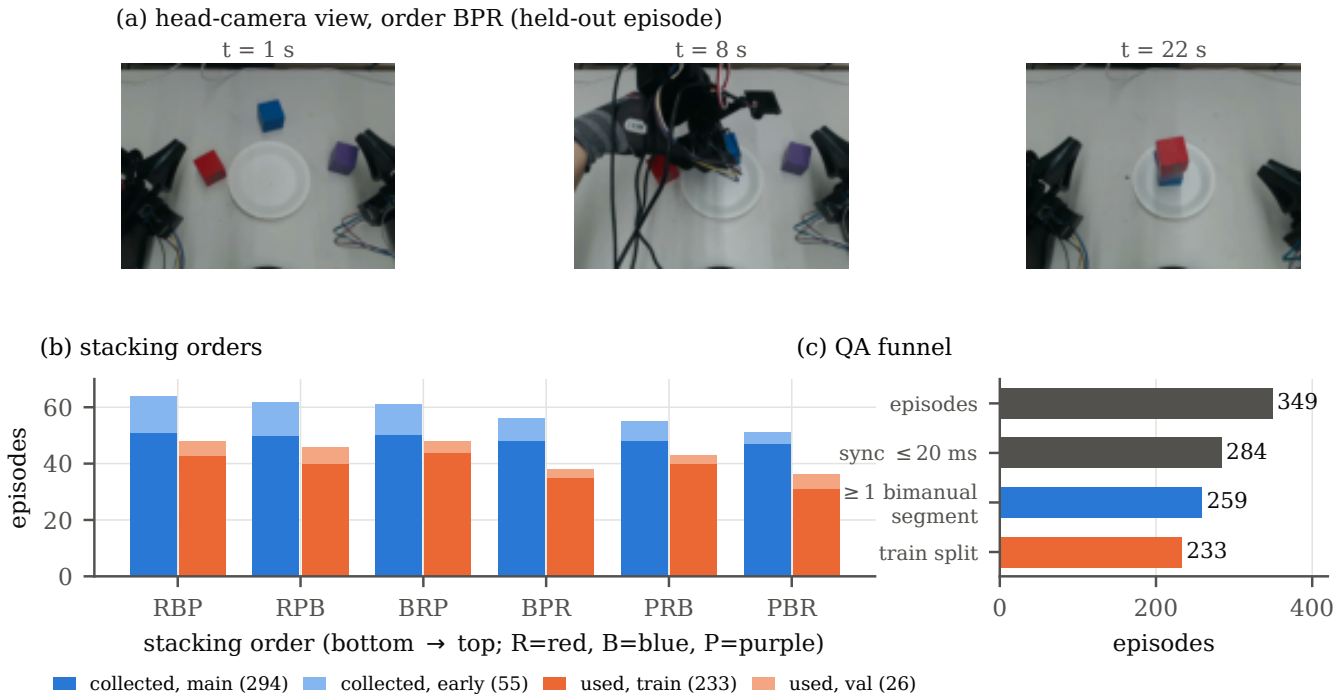


Fig. 1. Dataset design. (a) Head-camera frames at 1, 8 and 22 s of a held-out episode (order BPR). (b) Episodes per stacking order, as collected and after generation-1 processing. (c) Episode-level funnel.

TABLE I
OFFLINE HELD-OUT RESULTS, FINAL 300K CHECKPOINT (26 EPISODES). TCP ERROR: MEDIAN FK POSITION ERROR (MM); MAE: JOINT MAE (DEG), ZERO-MOTION PREDICTOR IN PARENTHESES.

Subset	Metric	k=1	k=8	k=30
All (798)	TCP err.	1.5	2.2	5.7
	MAE	0.30 (0.15)	0.42 (0.25)	0.97 (0.51)
Moving (187)	TCP err.	11.2	19.4	39.9
	MAE	2.02 (2.32)	3.29 (4.38)	6.59 (8.68)
	cosine	0.72	0.84	0.90

TABLE II
RETARGETING VARIANTS ON 1,229 CANDIDATE SEGMENTS, MEASURED AGAINST 120 HELD-OUT ROBOT EPISODES. NN: NEAREST-NEIGHBOUR JOINT DISTANCE TO ROBOT POSTURES (RAD, P50/P90).

Variant	Usable	NN p50 / p90	Start clusters
v1 single anchor	36.5%	1.49 / 1.55	1
v2-K multi-seed IK	66.6%	1.21 / 1.60	–
v2-Kr + posture penalty	67.0%	0.44 / 0.74	–
v2-TR window retrieval	56.1%	0.27 / 0.45	100

instruction's mean rank is 3.55 of 6 (chance 3.5): the model makes no task-useful use of the order.

Retargeting v2 and fine-tuning. The single start anchor was the main loss. Retrieving the closest feasible robot trajectory window (calibrated on 30 robot episodes, measured on 120 others) raises usable segments from 36.5% to 56.1% (Table II) and brings postures close to real robot postures (nearest-neighbour distance 1.49→0.27 rad, start clusters 1→100). The robot-like re-collection halved live wrist-rotation speed but raised usable segments only to 58.9%: workspace placement, not style, is the limit. Fine-tuned on 150 robot episodes, the ego-initialized model has a 10% lower geo score than one trained from the base model (95% CI 5.6–14.7%), but only on episodes inside both training sets, so this measures fit, not generalization.

IV. GENERATION 2: CARTESIAN EGO DATA

Contract. Joint retargeting removed a third to a half of the motion and used a different action space from our robot policies. Generation 2 converts each wrist trajectory directly into the robot data's representation: rows at 15 Hz; an action chunk of 16 relative poses $T(t)^{-1}T(t+k\Delta)$, $\Delta = 50.05$ ms (0.8 s), interpolated on the raw 30 Hz track; per arm position, 6-D rotation and gripper (20 dimensions, "CART20"), the rest of the 32-D head zeroed; state relative to each arm's task start. Source: 273 original and 60 HRL80 episodes, 330 after IMU-scale checks, with no workspace or IK filter.

Silent pose jumps. Ego states showed jumps above 100 mm that the robot data lacks. 1,482 of the 2,598 apparent jumps above 30 mm were time gaps between non-adjacent rows; the rest were real one-frame MASt3R-SLAM jumps without a tracking-loss flag, which corrupt every later start-anchored state. Truncating an episode after such a jump leaves 297 train

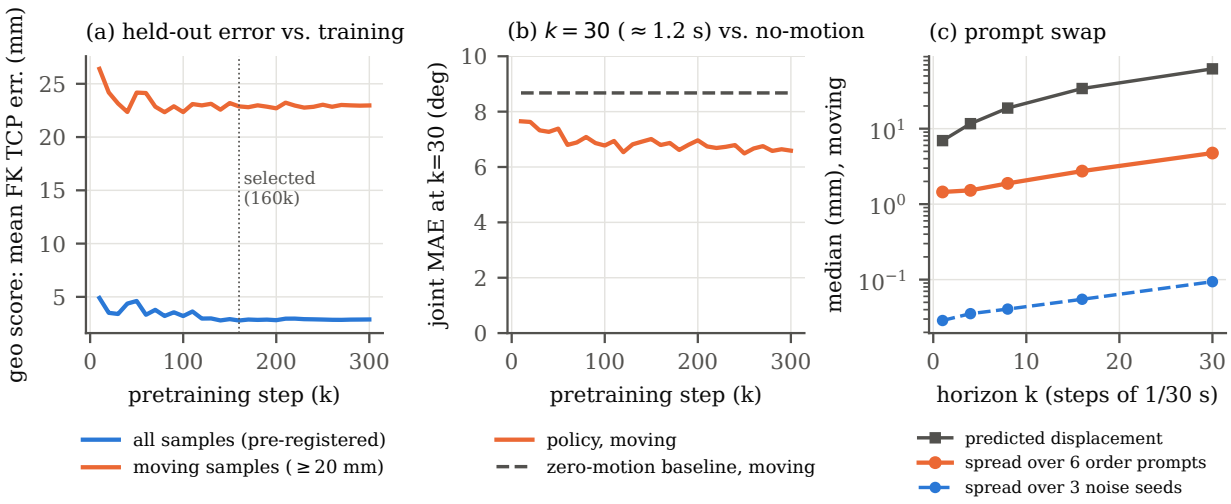


Fig. 2. Generation-1 offline evaluation. (a) Geo score over training, all versus moving samples. (b) Joint MAE at k=30 on moving samples, policy versus zero-motion predictor. (c) Prompt swap: spread across six instructions versus three noise seeds.

TABLE III
EGO-ONLY CHECKPOINT (40K), MEDIAN DIRECTION COSINE AT k=4/8/16 (LEFT | RIGHT) WHEN IMAGES AND STATE COME FROM DIFFERENT DOMAINS.

Images + state	Left k=4/8/16	Right k=4/8/16
ego + ego	0.87 / 0.91 / 0.92	0.89 / 0.94 / 0.84
ego + robot	0.76 / 0.81 / 0.90	0.72 / 0.69 / 0.72
robot + robot	0.36 / 0.30 / −0.01	0.19 / 0.02 / −0.15
robot + ego	0.16 / 0.24 / −0.16	0.49 / 0.33 / −0.03

episodes and 48,411 rows (−6.20%); the largest adjacent-row step falls from 422.5 to 90.5 mm.

Ego versus robot. The robot's IK reaches 99.7% (left) and 99.5% (right) of ego positions. Ego hands use a smaller workspace and are both still far more often (19.5% versus 2.4% of samples). The largest difference is visual: wrist-image sharpness (Laplacian variance) has a median of 1,854/1,487 for ego fisheye views and 59/64 for the robot's wrist cameras.

V. TRANSFER TO THE ROBOT

All runs use one recipe: X-VLA with loss on action dimensions 0–19, batch 8, 8-bit AdamW (policy learning rate 10^{-4}), bf16, seed 1000; fine-tuning uses robot normalization statistics. R312c is 312 teleoperated robot episodes of the same task.

Which soft prompt? X-VLA selects soft prompts by an embodiment id. In the public base weights only slots 10–17 were ever trained; the ids 0 and 6 we had used were untrained. All runs below use the unused slot 20.

The gap is in the wrist images. On 120 robot frames, an ego-only model (40k steps) has a median direction cosine at 0.4 s of −0.05/−0.03 (left/right), versus +0.81/+0.77 for a robot-trained reference; no axis or arm swap raises it above +0.16. Swapping inputs between domains (Table III), the ego model keeps a cosine of 0.69–0.90 with ego images whatever state it gets (robot state included), and falls to between −0.25 and +0.49 with robot images whatever the state. Re-rendering ego wrist frames as robot-like pinhole views (ROBOT100) lowers their sharpness to 181/132, still 2–3× the robot's; a 300k pretraining on this data finished on 5 October (training loss 0.007).

Initialization on 312 robot episodes. Fine-tuned on R312c from the base model (A) and from a 50k ego checkpoint (B) with identical data, schedule and seed, B starts at a 3.9× lower loss (0.945 for A and 0.244 for B at step 200); the ratio B/A is 0.49 over the first 5k steps, 0.875 over 5k–20k and 0.95 over 20k–40k, and the two are tied by 60k (0.0733 versus 0.0745; Fig. 3). Training loss only, one seed per arm; there is no held-out or closed-loop comparison of A and B.

Initialization on 30 robot episodes. To test the regime where human data should matter, we fine-tuned on R30 (R312c episodes 150–179, 17,600 frames, 600k-step schedule) from the base model and from the ROBOT100 pretraining at 100k and 300k steps (Table IV, Fig. 4). Both ego initializations start below half the scratch loss (window ratio 0.544 and 0.493 over 1k–5k); the 100k initialization reaches parity by about 100k steps (0.970 over 50k–100k, 1.021 over 100k–150k), the 300k one by about 200k. The 100k checkpoint is mid-schedule, and the 300k arm ran on a different GPU and PyTorch version, so the arms are not exact twins. All three were stopped at 471k, 305k and 310k steps after 139–214 epochs: every arm memorizes 30 episodes. No R30 checkpoint has been run on the robot.

On the robot. Between 2 and 4 October 71 checkpoints ran 49,431 control cycles, but the cycle log records IK tracking, not task outcome. The ego-initialized model fine-tuned on R312c for 300k (B300, checkpoints 75k–205k) completed the full three-cube stack in closed loop; trials were not counted and

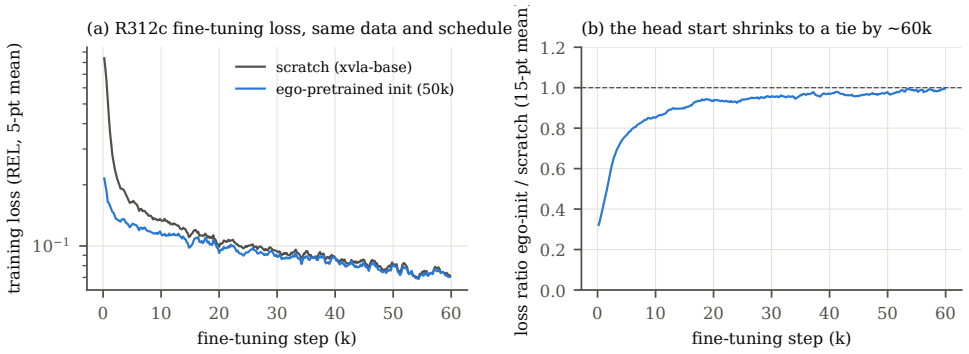


Fig. 3. R312c fine-tuning from the base model versus from the 50k ego checkpoint, same data, schedule and seed. (a) Training loss. (b) Ratio; values below 1 favour the ego initialization.

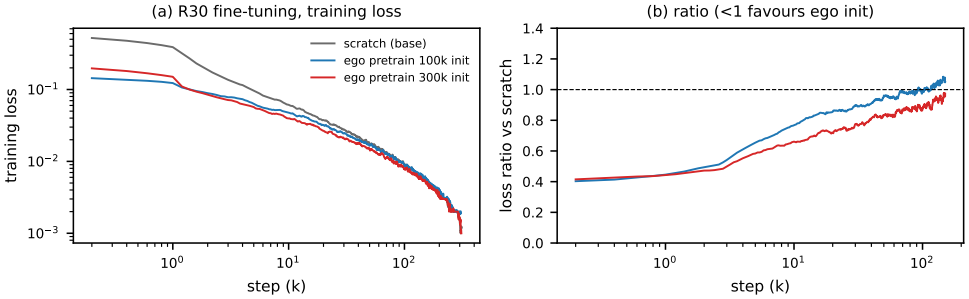


Fig. 4. R30 fine-tuning from the base model and from ego pretraining at 100k and 300k steps. (a) Training loss. (b) Ratio to scratch up to 150k. Training loss only.

TABLE IV
R30 FINE-TUNING: TRAINING LOSS AT MATCHED STEPS AND WINDOW-MEAN RATIO TO SCRATCH (<1 FAVOURS EGO). ONE SEED PER ARM.

Step / window	Scratch	Ego-PT 100k init	Ego-PT 300k init
200	0.801	0.239	0.446
10k	0.065	0.056	0.040
100k	0.009	0.009	0.007
ratio 1k–5k		0.544	0.493
ratio 50k–100k		0.970	0.856
ratio 100k–150k		1.021	0.914

no video was taken, so this is a demonstration with no success rate, and no scratch model on the same data was tested under the same protocol. Only the 150k and 200k B300 checkpoints survive a later storage clean-up. A trial logger now exists in the deployment interface but has recorded no trial.

VI. LESSONS, LIMITATIONS AND NEXT STEPS

(i) A pretrained model's conditioning interface must be audited: two of our ids pointed at untrained soft prompts. (ii) With a shared tensor contract, swapping inputs between domains localizes the gap, here to the wrist images. (iii) Pose and scale sources fail silently in some regime (one-frame SLAM jumps here, IMU scale on slow motion in Part II); an independent reference catches it. (iv) Training-loss comparisons of initializations end in a tie once robot data is memorized, at 312 and at 30 episodes alike; only held-out or closed-loop tests at early checkpoints can show a benefit.

Not supported by this study: a closed-loop success rate; any closed-loop difference between ego-initialized and scratch models; generalization to new layouts or unseen orders; transfer beyond one operator and one scene; transfer of the ego-only policy or the re-rendered images to robot cameras. **Next:** log every robot trial (checkpoint, layout, instruction, stack level 0–3, wrong colour, failure reason) with 20–30 trials per model; compare the R30 arms closed-loop at early and middle checkpoints (10k–150k); record layouts and hold out layout regions and two orders.

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